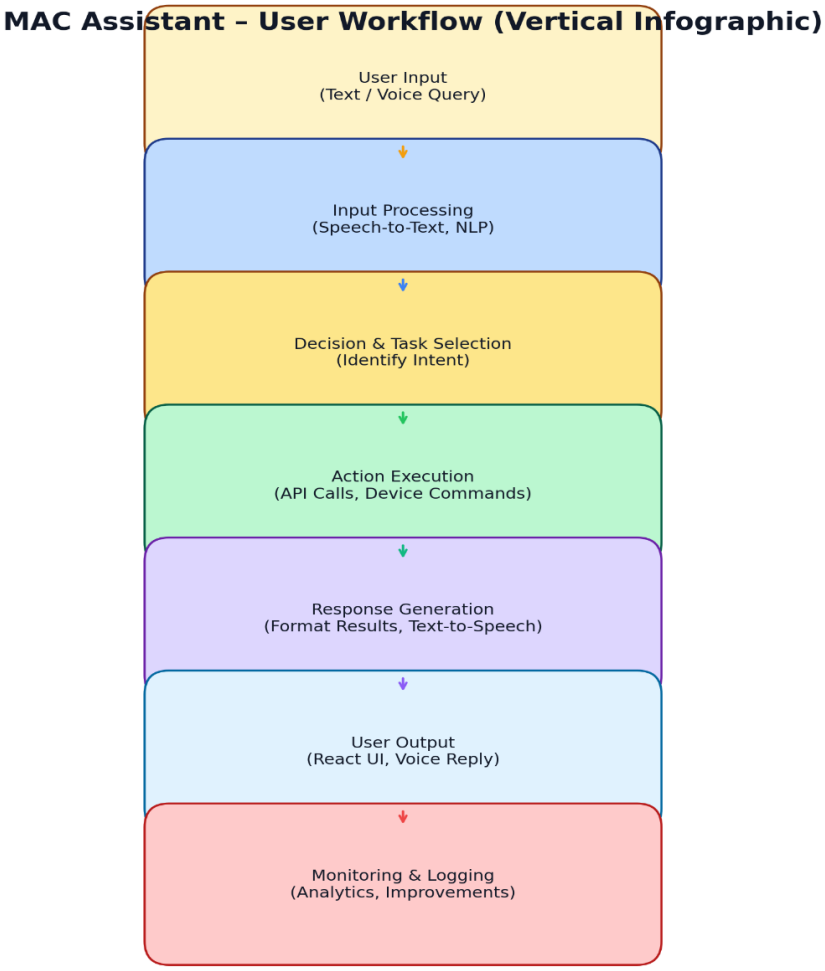
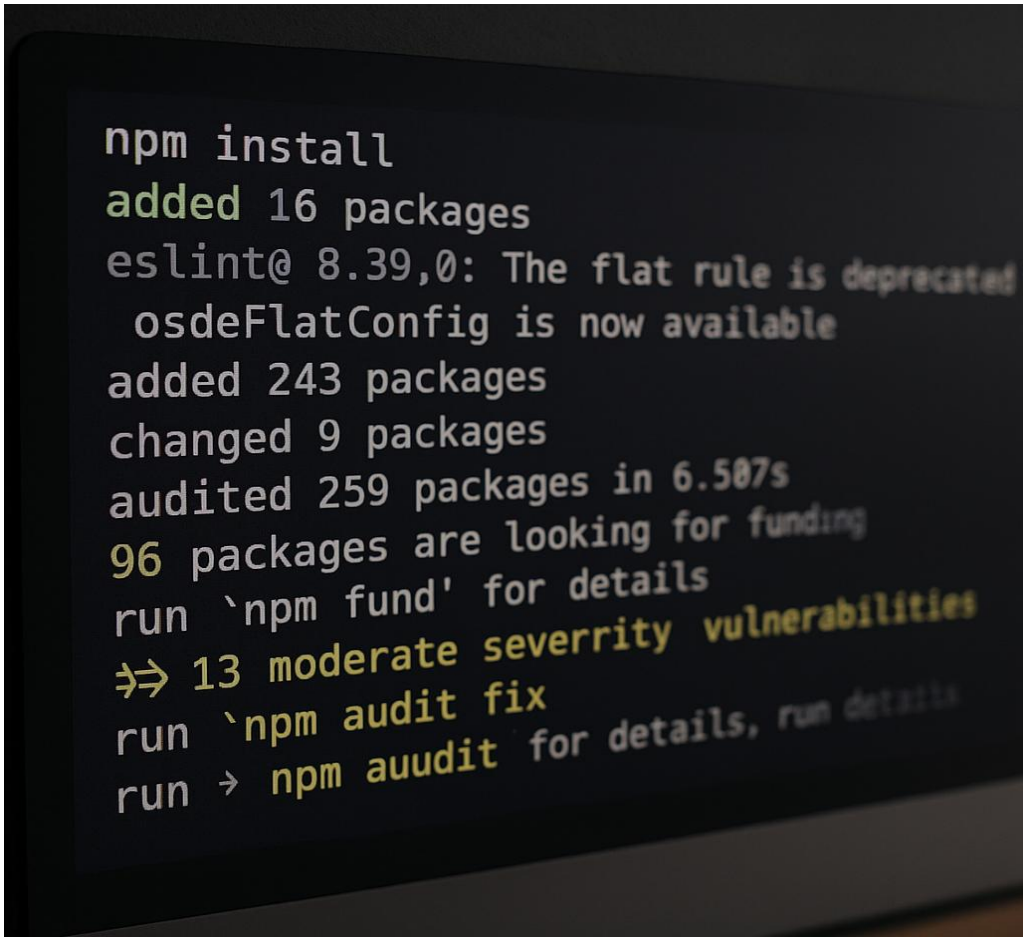
USER WORKFLOW (MAC -VOICE ASSISTANT)

1. **User Input**
    - User speaks or types a query into Siri Assistant.
2. **Input Processing**
    - Speech-to-Text (if voice).
    - NLP for intent recognition.
3. **Decision & Task Selection**
    - Identify what the user wants (search, app command, weather, reminder, etc.).
4. **Action Execution**
    - Perform API calls, trigger device functions, or run commands.
5. **Response Generation**
    - Process the results.
    - Convert text → speech if needed.
6. **User Output**
    - Display results in React UI (text, cards, visuals).
    - Provide voice response.
7. **Monitoring & Logging**
    - Store requests and responses for analytics & improvements.



Installations:


npm install
added 16 packages
eslint@ 8.39,0: The flat rule is deprecated
osdeFlatConfig is now available
added 243 packages
changed 9 packages
audited 259 packages in 6.507s
96 packages are looking for funding
run `npm fund' for details
13 moderate severrity vulnerabilities
run `npm audit fix
run npm auudit for details, run details

requirements.txt

siri_assistant_react > backend > requirements.txt

```
fastapi
uvicorn
speechrecognition
pyttsx3
pyaudio
```

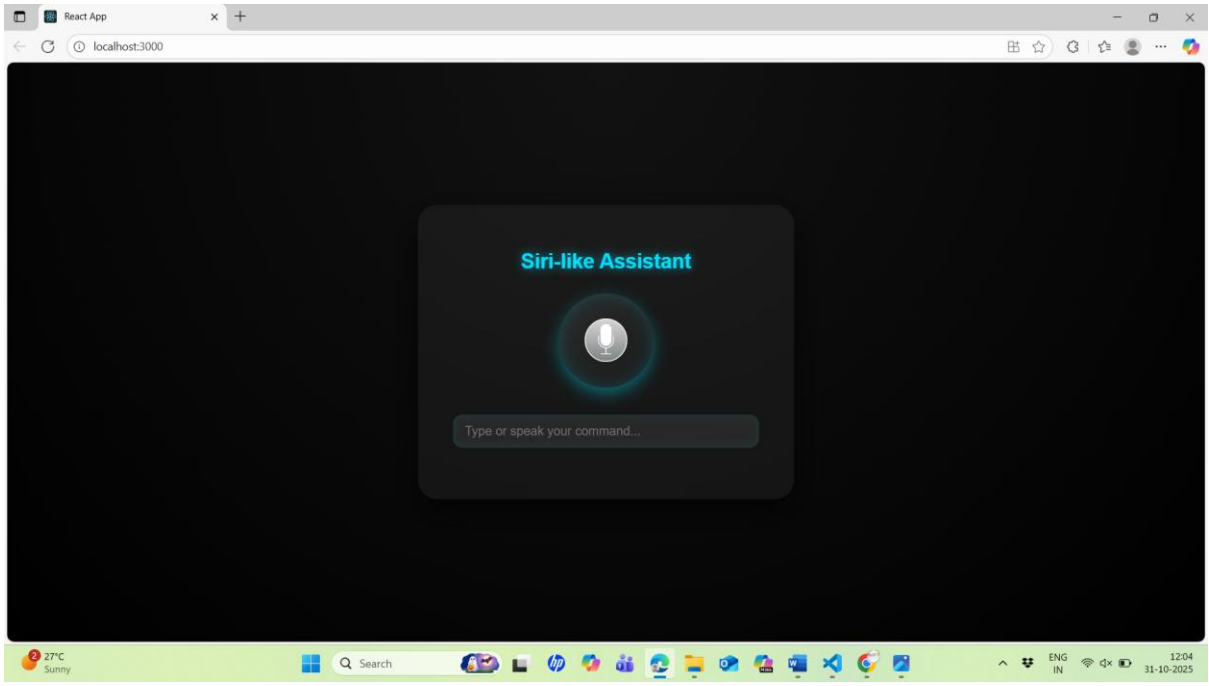
Project file structure:

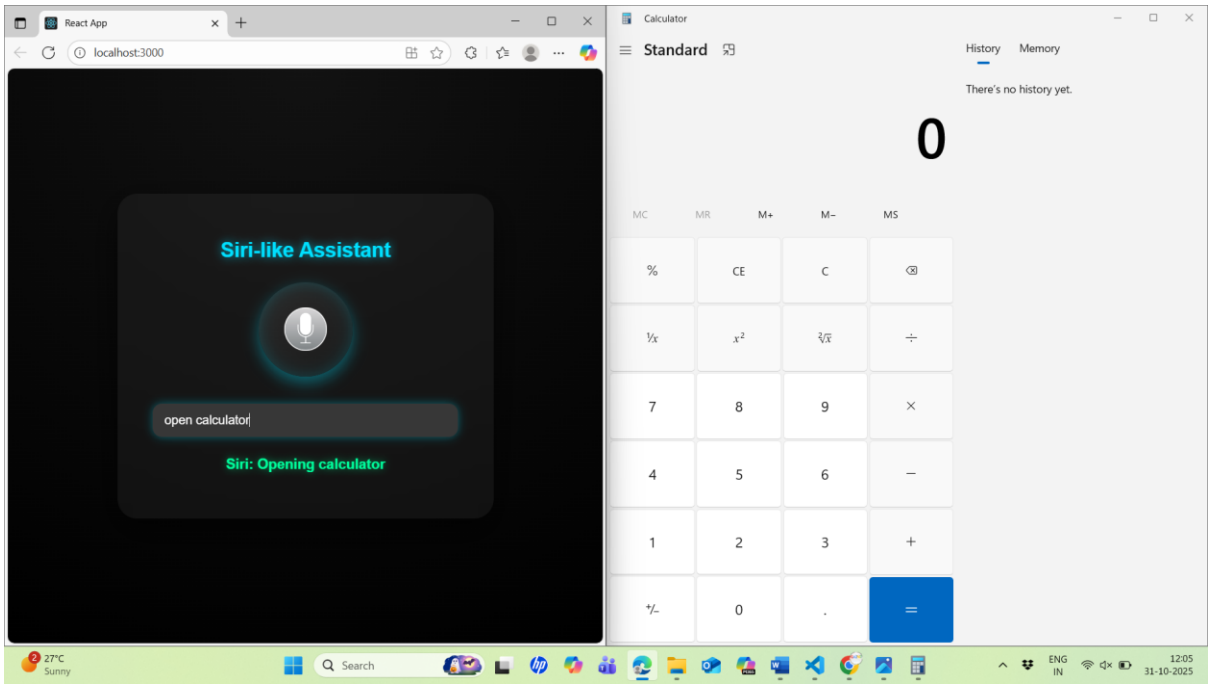
```
SIRI_ASSISTANT_REACT (1)
  siri_assistant_react
    .venv
    backend
      __pycache__
      venv
      app.py
      requirements.txt
    frontend
      node_modules
      public
        favicon.ico
        index.html
        logo192.png
        logo512.png
        manifest.json
        robots.txt
      src
        components
        App.css
        App.js
        index.js
        siri-mic.png
      .gitignore
      package-lock.json
      package.json
      README.md
    README.md
OUTLINE
TIMELINE
```

Outputs:

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\Gaint Works\siri_assistant_react (1)\siri_assistant_react\backend> flask run
ecify one.
PS E:\Gaint Works\siri_assistant_react (1)\siri_assistant_react\backend> uvicorn app:app
 --reload
INFO:     Will watch for changes in these directories: ['E:\\Gaint Works\\siri_assistant
_react (1)\\siri_assistant_react\\backend']
INFO:     Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO:     Started reloader process [1736] using WatchFiles
INFO:     Started server process [18084]
INFO:     Waiting for application startup.
INFO:     Application startup complete.
INFO:     127.0.0.1:58167 - "OPTIONS /process HTTP/1.1" 200 OK
INFO:     127.0.0.1:58167 - "POST /process HTTP/1.1" 200 OK
```

```
PROBLEMS    OUTPUT    DEBUG CONSOLE    TERMINAL    PORTS

PS E:\Gaint Works\siri_assistant_react (1)\siri_assistant_react\frontend> npm start

You can now view new in the browser.

  Local:            http://localhost:3000
  On Your Network:  http://10.183.20.212:3000

Note that the development build is not optimized.
To create a production build, use npm run build.

webpack compiled successfully
```


React App
localhost:3000
Siri-like Assistant
Type or speak your command...
27°C
Sunny
Search
ENG
IN
12:04
31-10-2025


React App
localhost:3000
Siri-like Assistant
open calculator
Siri: Opening calculator
Calculator
Standard
History
Memory
There's no history yet.
0
MC
MR
M+
M−
MS
%
CE
C
⅟x
x²
∛x
÷
7
8
9
×
4
5
6
−
1
2
3
+
+/−
0
.
=
27°C
Sunny
Search
ENG
IN
12:05
31-10-2025